PROBLEM 19

Heavy Particle on a Paraboloid

Original statement · Problems.pdf, p. 10

Paraboloid Surface

A heavy particle of mass m is moving without friction on the paraboloid which is described as:

$$az = r^2$$

where a is a positive constant and r is the distance to the Oz vertical axis. If $r = a$ initially, and the particle is released with a velocity v_0 tangent to the local parallel, reduce the problem to integrals of the form $\int [...] dr = \int [...] dt$. Determine the maximum and minimum of r along the motion.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m moves without friction on the paraboloid of revolution $az = r^2$, where $a > 0$ and $r = \sqrt{x^2 + y^2}$.
- **Degrees of Freedom:** 3 coordinates minus 1 constraint $\implies$ 2 CDOFs. We parametrize using cylindrical coordinates (r, θ) with $z = r^2/a$.
- **Initial State ($t = 0$):**
- Radius: $r(0) = a \implies z(0) = a^2/a = a$.
- Velocity: Launched tangent to the local parallel (pure azimuthal velocity):

$$\mathbf{v}_0 = v_0 \mathbf{e}_\theta \implies \dot{r}(0) = 0, \quad \dot{\theta}(0) = \frac{v_0}{a}, \quad \dot{z}(0) = 0$$

- **Parameters:** $m > 0$ [kg], $a > 0$ [m], $g > 0$ [m/s^2], $v_0 > 0$ [m/s].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Kinematics with $z = r^2/a$:**

$$\dot{z} = \frac{d}{dt}\left(\frac{r^2}{a}\right) = \frac{2r\dot{r}}{a}$$

Velocity vector:

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \frac{2r\dot{r}}{a}\mathbf{k}$$

Velocity squared:

$$v^2 = \dot{r}^2\left(1 + \frac{4r^2}{a^2}\right) + r^2\dot{\theta}^2$$

- **Change of basis (cylindrical basis).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is obtained from $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ by a rotation $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Surface-adapted basis.** With $s = \sqrt{1 + 4r^2/a^2}$, the unit tangent along the meridian is $\mathbf{e}_g = (\mathbf{e}_r + \frac{2r}{a}\mathbf{k})/s$ and the unit normal pointing into the bowl is $\mathbf{n}_{in} = (\mathbf{k} - \frac{2r}{a}\mathbf{e}_r)/s$, proportional to $-\boldsymbol{\nabla} f$ of Phase 3. Since $\mathbf{e}_g \times \mathbf{e}_\theta = \mathbf{n}_{in}$ the triple is right-handed, and

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2r/(as) \\ 0 & 1 & 0 \\ 2r/(as) & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1}{s^2} + \frac{4r^2}{a^2 s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- The determinants and products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

1. First Integral from the $\mathbf{e}_\theta$ Component of Newton's Law

Pedagogical Justification: The coordinate θ is cyclic (surface of revolution, vertical gravity). In cylindrical coordinates $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)$ (Notes Sec. 2.6.2). The normal to the surface $f(r,z) = r^2 - az = 0$ is $\boldsymbol{\nabla} f = 2r\mathbf{e}_r - a\mathbf{k}$, with no azimuthal component ($\mathbf{n}\cdot\mathbf{e}_\theta = 0$), and gravity acts along $-\mathbf{k}$. Hence the $\mathbf{e}_\theta$ equation of motion is $\frac{m}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0$.

$$r^2\dot{\theta} = r(0)^2\dot{\theta}(0) = a^2\left(\frac{v_0}{a}\right) = av_0$$

$$r^2\dot{\theta} = av_0 \implies \dot{\theta} = \frac{av_0}{r^2}$$

2. Conservation of Mechanical Energy

> **Pedagogical Justification:** The constraint is frictionless (bilateral holonomic), so the normal reaction does zero work. Gravity is conservative with potential $V = mgz = mg\frac{r^2}{a}$.

Initial energy:

$$E = \frac{1}{2}mv_0^2 + mgz(0) = \frac{1}{2}mv_0^2 + mga$$

Energy at time t:

$$E = \frac{1}{2}m\left[\dot{r}^2\left(1 + \frac{4r^2}{a^2}\right) + r^2\dot{\theta}^2\right] + mg\frac{r^2}{a} = \frac{1}{2}mv_0^2 + mga$$

Multiply by $\frac{2}{m}$:

$$\dot{r}^2\left(1 + \frac{4r^2}{a^2}\right) + r^2\left(\frac{av_0}{r^2}\right)^2 + \frac{2gr^2}{a} = v_0^2 + 2ga$$

$$\dot{r}^2\left(1 + \frac{4r^2}{a^2}\right) + \frac{a^2v_0^2}{r^2} + \frac{2gr^2}{a} = v_0^2 + 2ga$$

Grouping terms:

$$\dot{r}^2\left(1 + \frac{4r^2}{a^2}\right) + r^2\left(\dot{\theta}^2 + \frac{2g}{a}\right) = v_0^2 + 2ga$$

Compare with the quadrature formula in the official solution key (Problems.pdf, Solution keys section).

3. Reduction to Quadratures

Isolating $\dot{r}$:

$$\dot{r}^2 = \frac{v_0^2 + 2ga - \frac{a^2 v_0^2}{r^2} - \frac{2gr^2}{a}}{1 + \frac{4r^2}{a^2}} = \frac{r^2(v_0^2 + 2ga) - a^2 v_0^2 - \frac{2g}{a} r^4}{r^2\left(1 + \frac{4r^2}{a^2}\right)}$$

Taking the square root and separating variables:

$$\int \sqrt{\frac{1 + 4r^2/a^2}{v_0^2 + 2ga - a^2 v_0^2/r^2 - 2gr^2/a}}\, dr = \pm \int dt$$

4. Determination of Maximum and Minimum of r

Setting $\dot{r} = 0$:

$$v_0^2 + 2ga - \frac{a^2 v_0^2}{r^2} - \frac{2gr^2}{a} = 0$$

Multiply by ar^2:

$$ar^2(v_0^2 + 2ga) - a^3 v_0^2 - 2gr^4 = 0$$

$$2gr^4 - a(v_0^2 + 2ga)r^2 + a^3 v_0^2 = 0$$

Let $u = r^2$. The quadratic in u is:

$$2gu^2 - a(v_0^2 + 2ga)u + a^3 v_0^2 = 0$$

Since $r = a \implies u = a^2$ must be a root:

$$2g(a^2)^2 - a(v_0^2 + 2ga)(a^2) + a^3v_0^2 = 2ga^4 - a^3v_0^2 - 2ga^4 + a^3v_0^2 = 0$$

Factorizing $(u - a^2)$:

$$2gu^2 - a(v_0^2 + 2ga)u + a^3v_0^2 = (u - a^2)(2gu - av_0^2) = 0$$

The two roots for $u = r^2$ are:

$$u_1 = a^2 \implies r_1 = a$$

$$u_2 = \frac{av_0^2}{2g} \implies r_2 = v_0\sqrt{\frac{a}{2g}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Case 1:** $v_0 < \sqrt{2ga}$

$r_2 = v_0\sqrt{\frac{a}{2g}} < a$.

$$r_{max} = a, \qquad r_{min} = v_0\sqrt{\frac{a}{2g}}$$

The particle falls inwards towards the vertex, reaches r_{min}, and climbs back to a.

- **Case 2:** $v_0 = \sqrt{2ga}$

$r_2 = a$. Both roots coincide:

$$r_{min} = r_{max} = a$$

The particle moves on a horizontal circular orbit of constant radius $r = a$.

- **Case 3:** $v_0 > \sqrt{2ga}$

$r_2 = v_0\sqrt{\frac{a}{2g}} > a$.

$$r_{min} = a, \qquad r_{max} = v_0\sqrt{\frac{a}{2g}}$$

The particle climbs up the paraboloid to r_{max}, then falls back to a.

- Compare with the official solution key (Problems.pdf, Solution keys section).
- **Dimensional check:** $[v_0^2 + 2ga] = \mathrm{m}^2/\mathrm{s}^2$, $[a^2 v_0^2/r^2] = \mathrm{m}^2/\mathrm{s}^2$ and $[2gr^2/a] = \mathrm{m}^2/\mathrm{s}^2$; the roots $r_2 = v_0\sqrt{a/(2g)}$ have units $(\mathrm{m/s})\sqrt{\mathrm{m}/(\mathrm{m/s}^2)} = \mathrm{m}$.
- **Circular orbit (Case 2):** for $v_0^2 = 2ga$ the quadratic of Phase 3 becomes $2g\,u^2 - 4ga^2 u + 2ga^4 = 2g\,(u - a^2)^2$, a double root at $u = a^2$, so $\dot r \equiv 0$. This agrees with the circular-orbit condition $v^2 = g\,r\tan\gamma$ with $\tan\gamma = dz/dr = 2r/a = 2$ at $r = a$.
- **Limit $v_0 \to 0$:** $r_{min} = v_0\sqrt{a/(2g)} \to 0$, the particle drops to the vertex, as expected for a body released at rest.
- **Numerical check ($g = a = v_0 = 1$, case 1):** the turning points are the roots of $v_0^2 + 2ga - a^2 v_0^2/r^2 - 2gr^2/a = 0$; bisection gives $r_{min} = 0.707107$ against $v_0\sqrt{a/(2g)} = 0.707107$, and $r_{max} = a = 1$.

Source: Problems.pdf (Page 10)